Applying the Histogram of Oriented Gradients Algorithm for Detecting Grass Lay Direction

Ben Sunshine (Data Science) '24
Advisor: Professor Ivan Ramler (Statistics)

1. Motivation:

- Subsistence-oriented indigenous communities rely heavily on wind direction for hunting, fishing, and settlement
- Indigenous Traditional Ecological Knowledge (TEK) has observed a shift from historically predominant northerly wind patterns to southerly and easterly winds
- Due to lack of weather stations in remote areas of Alaska, we aim to use the Histogram of Oriented Gradients (HOG) algorithm to automate the measurement of grass lay direction conducted previously by Dr. Jon Rosales (Environmental Studies) to relate wind direction to grass lay direction

2. Data:

- Acquired various images from Google Images (diagonal lines, San Francisco, grass) to assess the algorithm's abilities
- Applied to an aerial grass image from St. Lawrence University's Living Lab as a proxy for Alaskan Tundra Grasses

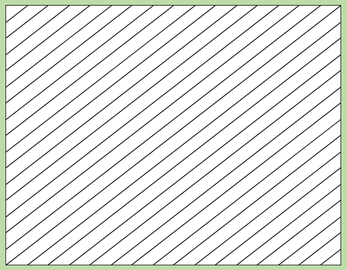


Image 1: Diagonal Lines

Image 2: Downtown San Francisco, CA

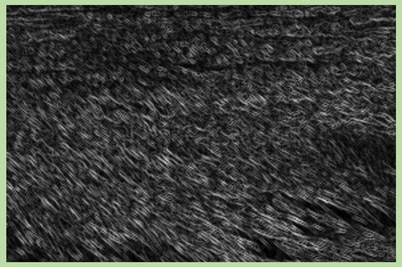
Image 3: Internet Image of Grass

Image 4: SLU Living Lab Weather Station

3. Methods:

- Convert image to grayscale then proportionality rescale the image
- Calculate magnitude and angle of gradient of each pixel:

$$G_x = I(r, c+1) - I(r, c-1)$$
$$G_y = I(r+1, c) - I(r-1, c)$$
$$Magnitude(\mu) = \sqrt{G_x^2 + G_y^2}$$
$$Angle(\theta) = tan^{-1}(\frac{G_y}{G_x})$$

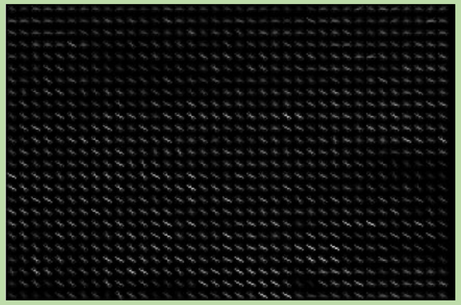


- Create a histogram of gradient magnitudes and angles:

121	10	78	96	125
48	152	68	125	111
145	78	85	89	65
154	214	56	200	66
214	87	45	102	45

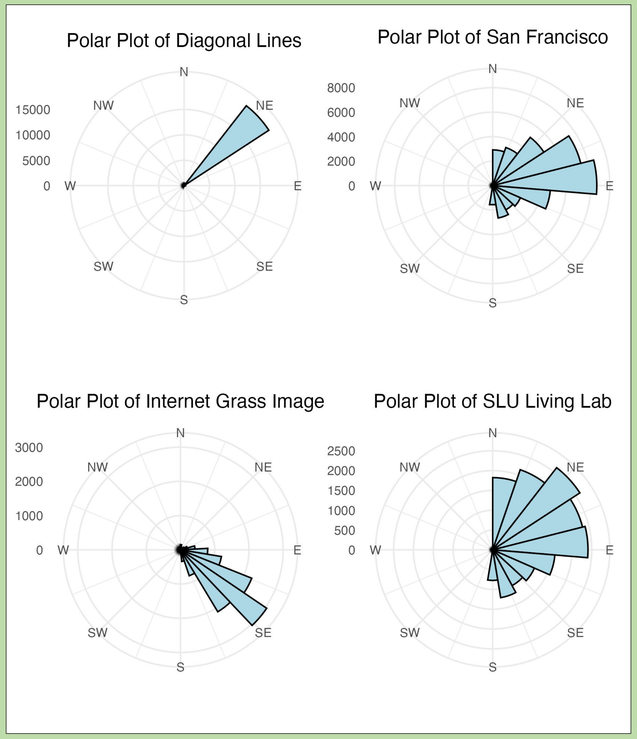
Frequency						1									
Angle	1	2	3	4 ...	35	36	37	38	39....	175	176	177	178	179	180

Histogram of Oriented Gradients Algorithm. Analytics Vidhya, https://www.analyticsvidhya.com/blog/2019/09/feature-engineering-images-introduction-hog-feature-descriptor/.



- Create polar plots of angles and compare to original images

4. Results:



5. Conclusion:

- Polar plots successfully identified visual trends from images
- Biggest factor in getting an accurate polar plot is image quality
- Future Analysis:
  - Compare the hand measured grass lay angles with those generated by the HOGs algorithm.
  - Acquire the wind data from the Living Lab weather station to see how the predominant wind direction correlate with grass lay direction generated by the HOGs algorithm
  - Use drones to take images to determine if wind direction has changed from predominantly northern winds